

Task 3

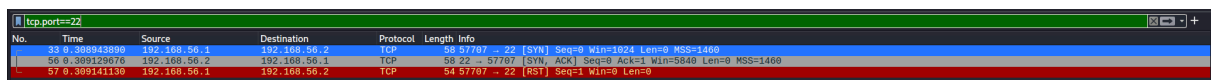
In this task we'll do some TCP, UDP Scan via `nmap`

TCP Scan

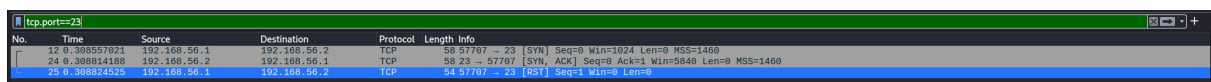
```
(ammar@no0ne)-[~]
$ sudo nmap -sS -p 21,22,25 192.168.56.2
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-03 14:29 +03
Nmap scan report for 192.168.56.2
Host is up (0.00027s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    open  smtp
MAC Address: 08:00:27:CB:51:F5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.26 seconds
```

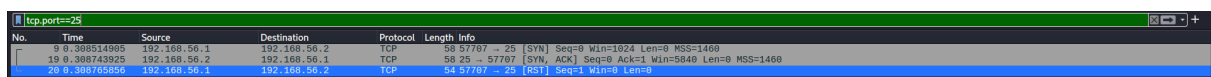
- `-sS` stands for Stealth Scan, This scan doesn't establish a connection, after receiving the `SYN,ACK` packet it sends `RST` packet to close the connection
- `-F` Scan the first 1,000 famous ports



No.	Time	Source	Destination	Protocol	Length	Info
34	0.308843080	192.168.56.1	192.168.56.2	TCP	58	57767 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
57	0.309141130	192.168.56.1	192.168.56.2	TCP	54	57767 → 22 [RST] Seq=1 Win=0 Len=0



No.	Time	Source	Destination	Protocol	Length	Info
12	0.308557021	192.168.56.1	192.168.56.2	TCP	58	57767 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
25	0.308824525	192.168.56.1	192.168.56.2	TCP	54	57767 → 23 [RST] Seq=1 Win=0 Len=0



No.	Time	Source	Destination	Protocol	Length	Info
9	0.308514905	192.168.56.1	192.168.56.2	TCP	58	57767 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
20	0.308743925	192.168.56.1	192.168.56.2	TCP	54	57767 → 25 [RST] Seq=1 Win=0 Len=0

A normal TCP connection would be:

1. SYN
2. SYN, ACK
3. ACK

If a device on a specific port accepts this connection the port would be considered open, So on nmap the tools uses this connection, sends the `syn` packet wait for the `syn, ack` packet then sends a `rst` packet (reset packet) this packet terminates the connection. Thus, from the scanned host prespective the connection wasn't established so it isn't logged which prevent detection.

UDP Scan

```
(ammar@noOne)-[~]
$ sudo nmap -sU -p 21,22,25 192.168.56.2
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-03 14:29 +03
Nmap scan report for 192.168.56.2
Host is up (0.00032s latency).

PORT      STATE SERVICE
21/udp    closed ftp
22/udp    closed ssh
25/udp    closed smtp
MAC Address: 08:00:27:CB:51:F5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds
```

No.	Time	Source	Destination	Protocol	Length	Info
225	16.596055317	192.168.56.1	192.168.56.2	UDP	42	50770 - 25 Len=0
226	16.596075223	192.168.56.1	192.168.56.2	UDP	42	50770 - 21 Len=0
227	16.596082075	192.168.56.1	192.168.56.2	UDP	42	50770 - 22 Len=0

- **-sU** Scan works by sending raw UDP packets to target ports and analyzing the response (or lack of response) from the remote host
Because UDP is **connectionless**, there is no virtual connection or handshake established.

Service Identification

```
(ammar@noOne)-[~]
$ sudo nmap -sS -sV -p 21,22,25 192.168.56.2
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-03 14:29 +03
Nmap scan report for 192.168.56.2
Host is up (0.00032s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
25/tcp    open  smtp     Postfix smtpd
MAC Address: 08:00:27:CB:51:F5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: metasploitable.localdomain; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
```

- **21** FTP, File Transfer Protocol
- **22** SSH, Secure Shell Host
- **25** SMTP, Simple Mail Transfer Protocol